

MBA IN SOFTWARE ENGINEERING

Domain-driven Designer

Guilherme Lima

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Domain-Driven Designer

Professor Guilherme Lima

Schedule

01

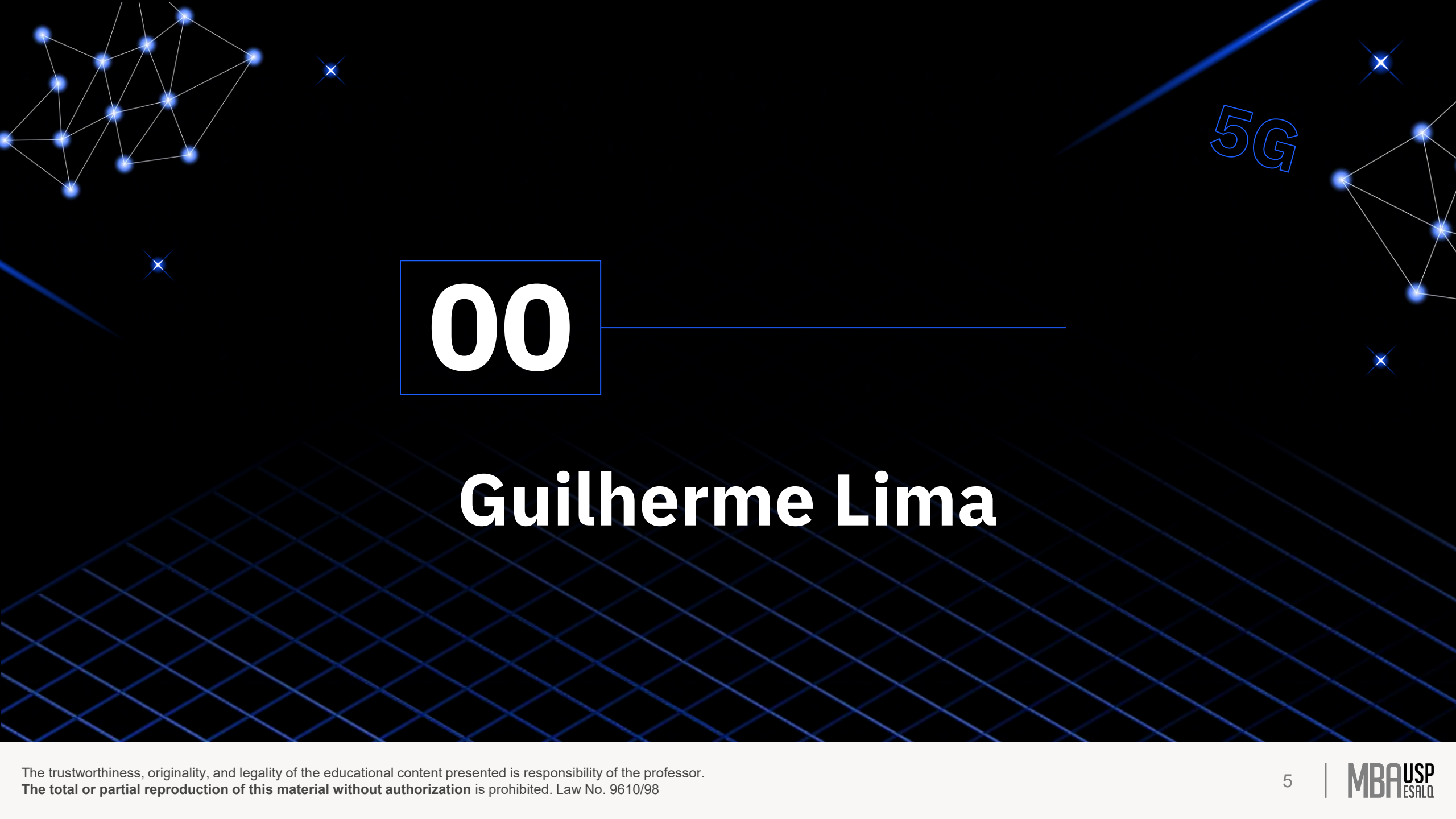
DDD and its impact on business alignment

02

Case studies

03

Ubiquitous Language



00

Guilherme Lima





Building software is more
challenging than ever.

Some reasons

1

Business complexity

The software system needs to meet specific demands

2

Perfect systems

The market demands flawless interfaces, high performance, and 24/7 availability

3

Technological diversity

The variety of available languages, frameworks, and tools makes choosing the right technologies a challenge

Artificial Intelligence



" data-bbox="42 37 919 901"/>

GITHUB COPILOT: CHAT

GitHub Copilot

Hi @monalisa, how can I help you?

I'm powered by AI, so surprises and mistakes are possible. Make sure to verify any generated code or suggestions, and share feedback so that we can learn and improve.

Ask a question or type '/' for commands

parse_expenses.py × addresses.rb × sentiments.ts ×

```
1 import datetime
2
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```

TS button.ts

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16

Accept

Discard





Constantly changing technological trends

An online streaming service

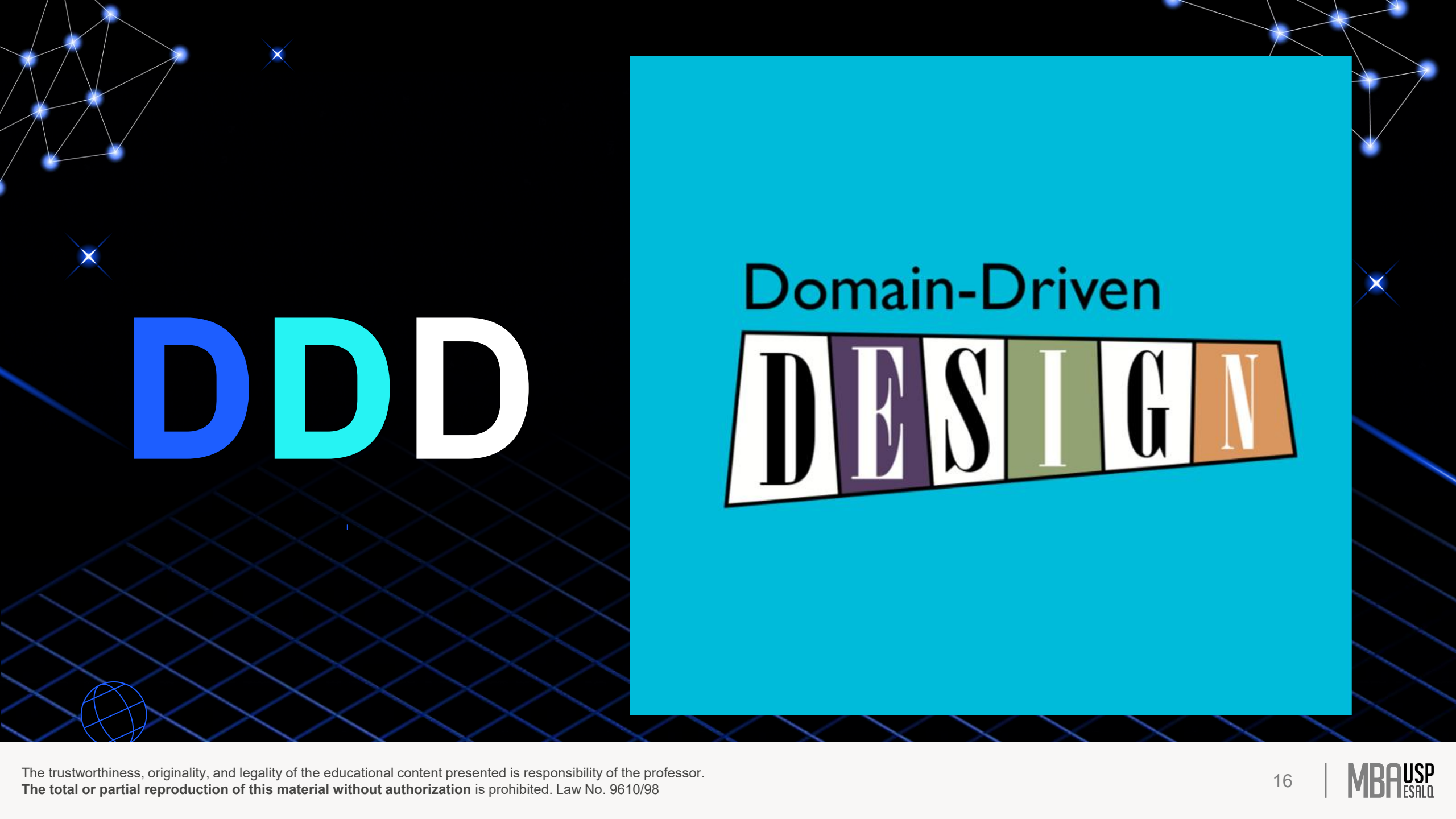
Company	Technologies	Customers
Netflix	AWS, Kubernetes, Java, Python	247 million
Amazon Prime	AWS, Java, React, Python	200 million
Disney+	AWS, Node.js, React, JavaScript	146 million

An online streaming service

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Our company 		

An online streaming service

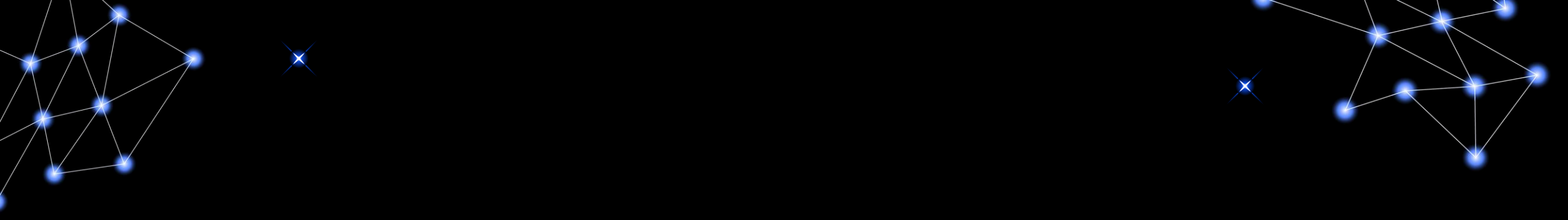
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Our company 🤝	AWS, Java, Python	Millions of customers?



DDD

Domain-Driven

DESIGN



✦ Create systems that **accurately** ✦
reflect the *specific rules, processes,*
and languages of the domain in
which the software will be applied.



Every software program relates to some activity or interest of its user.

Software *domain*

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Software *domain*

Every software program relates to some activity or interest of its user.

intangibles



Physic



Valuable software
is aligned with an
engineering team that
understands the
business
activity.

✦ **Valuable software**
is aligned with an
engineering team that
understands the
business
activity.





01

DDD and its impact on business alignment



Clear code always
seems to have
been written by
someone who
cares.

Robert C. Martin (Uncle Bob)



*DDD prioritizes **collaboration**
between **developers** and domain
experts.*



Business Domains

A **business domain** defines the main area of activity of a company.





NETFLIX



Disney+

Microsoft



TIM

YouTube

Spotify®



SAMSUNG

Uber

airbnb

LATAM

SAP







NETFLIX



NETFLIX
Entertainment

Communication

Gmail

NETFLIX
Entertainment

Social Media

Disney+
Entertainment

Microsoft
Technology

hardware
NVIDIA

TIM
telecommunications

YouTube
Videos

Spotify®
Streaming

Apple
Technology

SAMSUNG
Technology

Uber
Mobility

airbnb
Rentals

LATAM
Transport

SAP
management systems

Service the
company *provides*
to its customers





Subdomains

NETFLIX

Main domain
Digital streaming entertainment

NETFLIX

Main domain
Digital streaming entertainment

**Content
Catalog**

NETFLIX

Main domain
Digital streaming entertainment

**Content
Catalog**

Indexing

NETFLIX

Main domain
Digital streaming entertainment

**Content
Catalog**

Indexing

**Rights
management**

NETFLIX

Main domain
Digital streaming entertainment

**Content
Catalog**

Indexing

**Rights
management**

Infrastructure

NETFLIX

Main domain
Digital streaming entertainment

**Content
Catalog**

Indexing

**Rights
management**

Infrastructure

**Recommendation
Algorithms**

NETFLIX

Main domain
Digital streaming entertainment

**Content
Catalog**

Indexing

**Rights
management**

Infrastructure

**Recommendation
Algorithms**

**Monetization and
Subscription**

Domain experts

Domain experts **are**
not the **analysts** who
gather requirements nor
the **engineers** who
design the system.





*Do A,
okay?*

*And what
about B, we
don't do it?*

*So we
don't do
A?*

*Doing C, I'll
deliver it
today!*

Communication

Ubiquitous Language

Ubiquitous Language

One of the pillars of DDD

It's a shared vocabulary between developers and domain experts.

Eliminates ambiguities

Developers, business analysts, and stakeholders, use the same terms with the same meanings.

Improves communication

Reduces misunderstandings and ensures that everyone understands and works according to the same domain concepts.



PROBLEM

In the system, there is a function called **Product Availability**, but the term is not well-defined among the teams.

Product Availability

Business team

Products that can be sold, considering completed sales and those in the purchasing process.

Development team

Focuses on the physical quantity of products in stock that can be sold

Domain Expert

I understand an item's **ability to be acquired**, taking into account both **actual stock and items reserved in the shopping cart**

What now?

6 practical actions to implement a **ubiquitous language**

Documents

01

02

Regular Meetings

**Meaningful Names in
Code**

03

04

Alignment

Internal Wiki

05

06

**Review and
Refactor**

Case Study

Thank you!

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MBA **USP**
ESALQ